

Convergence to Seasoned Equity Offering Market Efficiency

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Abstract

The paper examines relation between returns and contemporaneous as well as lagged order imbalances by regression. The empirical result shows that contemporaneous imbalance have a significantly positive impact on return. Besides, conditional on contemporaneous imbalances, there is a significantly negative relation between lagged-one imbalance and return, except for ten minutes interval. It implies that SEO market converge to efficiency within ten minutes.

In addition, we examine dynamic relation between volatility and order imbalances by a GARCH model. The empirical results show that market makers mitigate volatility well in SEO announcements. Market makers do have the capacity to stabilize the price in secondary market making. Finally, we develop an imbalance based trading strategy to test market efficiency. We use quote and trade prices. We find that trade price based earns a positive return, and beats the original open-to-close return. A nested causality approach, which examines the dynamic return-order imbalance relationship during the price formation process, confirms the results.

1. Introduction

Spiess and Affleck-Graves (1995) document that firms making seasoned equity offerings during 1975-1989 substantially underperformed a sample of matched firms from the same industry and of similar size that did not issue equity. It implies that SEO firms virtually do not have good performance. And managers take advantage of overvaluation in seasoned equity offering markets. Teoh, Welch, and Wong (1998) find that SEO firms report higher earnings by changing the financial statement, but in fact, they do not have that good performance. Rangan (1998) finds that earnings management around the time of the offering explains a portion of the poor performance. Altinkilic and Hansen (2003) find that expected discounting in seasoned equity offers is a cost of uncertainty about firm value, marketing new shares, and acquiring information that raises the offer price. Carlson, Fisher, and Giammarino (2006) present a rational theory of SEOs that explains a pre-issuance price run-up, a negative announcement effect, and long-run post-issuance underperformance.

Chordia and Subrahmanyam (2004) document that order imbalances, even lagged five days, explain return. In this study, we examine the convergence process to market efficiency of seasoned equity offering firms. We are interested in price formation process in which seasoned equity offering news to incorporate into stock prices. We examine the relationship between order imbalances and return by regression. We employ a time varying GARCH(1,1) to investigate the relation between volatility and order imbalance. In order to examine the convergence process, we choose 5 minutes, 10 minutes, and 15 minutes time intervals. Barclay and Warner(1993) find out that most of cumulative stock price change is due to medium-size orders. This result is consistent with the hypothesis that informed traders tend to separate their orders into medium-size orders to conceal their private-owned information, and thus stock price movement is mainly due to informed traders' private information.

Furthermore, we develop an imbalance based trading strategy in different time intervals. And we expect to earn a positive return and even beat the original open-to-close return. Finally, we investigate the nested causality between order imbalances and returns as we explore the intraday dynamics that is essential in convergence process.

The remainder of this paper is organized as follows. Section 2 describes data and methodology. Section 3 presents the empirical results, and Section 4 concludes.

2. Data and Methodology

We select seasoned equity offering firms from CRSP (Center for Research in Security Prices) from January 1, 2004 to December 31, 2004. We obtain intraday trading data from TAQ (Trade and Automated Quotations). All samples meet the following criteria:

1. The sample shall be included in both the CRSP and TAQ.
2. If there are any stock splits, reverse splits, stock dividends, or repurchases, the firm will be deleted from our sample.
3. We only focus on the trading period during 9:30AM to 4:00PM, so any trade that records before open or after closing is ignored.
4. Any quote less than five seconds prior to the trade is ignored and the first one at least five seconds prior to the trade is retained.

We use the Lee and Ready (1991) trade assignment algorithm to estimate buyer-initiated or seller-initiated order. Then, we calculate returns and order imbalances for three different time intervals, namely 5, 10, and 15 minutes. We collect 97 SEO firms in our sample. The average return in our sample is -0.52%, with a median -0.46%. The standard deviation of return is 2.08%, with a maximum of 4.58% and a minimum of -7.24. First of all, we examine unconditional return-order imbalance regression relation.

$$R_t = \alpha_0 + \alpha_1 OI_{t-1} + \alpha_2 OI_{t-2} + \alpha_3 OI_{t-3} + \alpha_4 OI_{t-4} + \alpha_5 OI_{t-5} + \varepsilon_t, \quad (1)$$

where R_t is the current stock return of the individual stock on time t ,

OI_{t-i} , $i=1, 2, 3, 4$, and 5 , are lagged order imbalance at time $t-1$, $t-2$, $t-3$, $t-4$, and $t-5$ of each individual stock,

α_0, α_i , $i=1, 2, 3, 4$, and 5 , are intercepts and coefficients of the corresponding lagged order imbalance,

and ε_t is the error term.

We expect a significantly positive relation between current returns and lagged order imbalances. Then, we investigate conditional return-order imbalance relation with contemporaneous order imbalance included for three different time intervals. We anticipate

By using the coefficients and t -values of the order imbalances, we want to find the relation between the interval returns and current as a significantly positive contemporaneous return-order imbalance and a negative lag order imbalances relation because of “information overweighting”, propose by Chordia and Subrahmanyam (2004).

We employ a time varying GARCH(1,1) model to investigate dynamic relation between volatility and order imbalance.

$$R_t = \alpha + \varepsilon_t$$

$$\varepsilon_t | \Omega_{t-1} \sim N(0, h_t)$$

$$h_t = A + B h_{t-1} + C \varepsilon_{t-1}^2 + \gamma * OI_t \quad (3)$$

where R_t is the return in period t , and is defined as $\ln\left(\frac{P_t}{P_{t-1}}\right)$,

OI_t is the explanatory variable, order imbalance,

γ is the coefficient describing the impact of order imbalance on stock volatility,

ε_t is the residual value of the stock return in period t ,

Ω_{t-1} is the information set in period $t-1$,

and α , A , B , and C are intercepts and coefficients.

We expect a significantly positive coefficient γ , which represents the relation

between OI and volatility, because a large volatility is followed by a large order imbalance intuitively.

Finally, in order to explain the story behind an imbalance-based trading strategy, we employ a nested causality to explore the dynamic causal relationship between returns and order imbalances. According to Chen and Wu (1999), we define four relationships between two random variables, x_1 and x_2 , in terms of constraints on the conditional variances of $x_{1(T+1)}$ and $x_{2(T+1)}$ based on various available information sets, where $x_i = (x_{i1}, x_{i2}, \dots, x_{iT})$, $i=1, 2$, are vectors of observations up to time period T .

Definition 1: Independency, $x_1 \perp x_2$:

x_1 and x_2 are independent if

$$Var(x_{1(T+1)} | \underset{\sim}{x_1}) = Var(x_{1(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}) = Var(x_{1(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}, \underset{\sim}{x_{2(T+1)}}) \quad (4)$$

and

$$Var(x_{2(T+1)} | \underset{\sim}{x_2}) = Var(x_{2(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}) = Var(x_{2(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}, \underset{\sim}{x_{1(T+1)}}) \quad (5)$$

Definition 2: Contemporaneous relationship, $x_1 \longleftrightarrow x_2$:

x_1 and x_2 are contemporaneously related if

$$Var(x_{1(T+1)} | \underset{\sim}{x_1}) = Var(x_{1(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}) \quad (6)$$

$$Var(x_{1(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}) > Var(x_{1(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}, \underset{\sim}{x_{2(T+1)}}) \quad (7)$$

and

$$Var(x_{2(T+1)} | \underset{\sim}{x_2}) = Var(x_{2(T+1)} | \underset{\sim}{x_1}, \underset{\sim}{x_2}) \quad (8)$$

$$Var(x_{2(T+1)} \Big|_{\sim} x_1, x_2) > Var(x_{2(T+1)} \Big|_{\sim} x_1, x_2, x_{1(T+1)}) \quad (9)$$

Definition 3: Unidirectional relationship, $x_1 \Rightarrow x_2$:

There is a unidirectional relationship from x_1 to x_2 if

$$Var(x_{1(T+1)} \Big|_{\sim} x_1) = Var(x_{1(T+1)} \Big|_{\sim} x_1, x_2) \quad (10)$$

and

$$Var(x_{2(T+1)} \Big|_{\sim} x_2) > Var(x_{2(T+1)} \Big|_{\sim} x_1, x_2) \quad (11)$$

Definition 4: Feedback relationship, $x_1 \Leftrightarrow x_2$:

There is a feedback relationship between x_1 and x_2 if

$$Var(x_{1(T+1)} \Big|_{\sim} x_1) > Var(x_{1(T+1)} \Big|_{\sim} x_1, x_2) \quad (12)$$

and

$$Var(x_{2(T+1)} \Big|_{\sim} x_2) > Var(x_{2(T+1)} \Big|_{\sim} x_1, x_2) \quad (13)$$

To explore the dynamic relationship within a bi-variate system, we form the five statistical hypotheses in Table 1 where the necessary and sufficient conditions corresponding to each hypothesis are given in terms of constraints on the parameter values of the VAR model.

To determine whether there exists a specific causal relationship, we use a systematic multiple hypotheses testing method. Unlike the traditional pair-wise hypothesis testing approach, this testing method avoids the potential bias induced by restricting the causal relationship to a single alternative hypothesis. To implement this method, we employ the results of several pair-wise hypothesis tests. For instance, in order to conclude that $x_1 \Rightarrow x_2$,

we need to establish that $x_1 < \neq x_2$ and to reject that $x_1 \neq > x_2$. To conclude that $x_1 < - > x_2$, we need to establish that $x_1 < \neq x_2$ as well as $x_1 \neq > x_2$ and also to reject $x_1 \wedge x_2$. In other words, it is necessary to examine all five hypotheses in a systematic way before we draw the conclusion that a dynamic relationship exists. The following presents an inference procedure that starts from a pair of the most general alternative hypotheses.

Our inference procedure for exploring the dynamic relationship is based on the principle that a hypothesis should not be rejected unless there is sufficient evidence against it. In the causality literature, most tests intend to discriminate between independency and an alternative hypothesis. The primary purpose of the literature cited above is to reject the independency hypothesis. On the contrary, we intend to identify the nature of the relationship between two financial series. The procedure consists of four testing sequences, which implement a total of six tests (denoted as (a) to (f)), where each test examines a pair of hypotheses. The four testing sequences and six tests are summarized in a decision-tree flow chart in Table 2.

3. Empirical Results

3.1 Unconditional Lagged Return-Order Imbalance Regression Relation

We use a regression to examine unconditional lagged return-order imbalance relation. The results are exhibited in Table 3. At 5% significant level, significantly negative percent of lagged-one imbalance are all the same 4.12% for 5, 10, and 15 minutes intervals. At 1% significant level, the significantly negative percent of lagged-one imbalance are only 0%, 1.03%, and 3.09% for 5, 10, and 15 minutes intervals respectively. The coefficients of lagged-one order imbalance in 5 and 15 minutes interval are negative, but to our surprise, the

coefficient of lagged-one imbalance in 10 minutes interval is positive, which are inconsistent with Chordia and Subrahmanyam (2004). The possible explanation is as follow. SEO firms overestimate firm values while the public is not a fool. They eager to sell, herding or spreading their orders once the firm announces SEO, which causes a huge negative order imbalance. Confronted by the imbalance pressure, within 5 minutes, market makers overreact. However, in 10 minutes, market makers are able to mitigate price volatility. They lower the quote continuously, and makes investors who still sell the SEO stocks earn negative 10 minutes return. Though, from a daily perspective, if the information is strong enough, market makers do not have the ability to shield the overwhelming selling orders. This is why the coefficient of lagged-one imbalance in 15 minutes is negative.

In addition, we find a decreasing pattern from 5 to 15 minutes interval, although the total significant percentage of lagged-one declines first at the 10 minutes interval and then goes up at the 15 minutes interval at all significant levels. We still observe a convergence pattern from 5 to 15 minutes from 7.22% to 6.19% at the 10% significant level.

3.2 Conditional Contemporaneous Return-Order Imbalance Regression Relation

Table 4 shows that contemporaneous order imbalance are significantly positive in all significant levels and time intervals. And most of the coefficients of lagged imbalances are significantly negative, which is consistent with Chordia and Subrahmanyam (2004). According to Chordia and Subrahmanyam (2004), when market makers confront a positive (negative) contemporaneous imbalance, they raise (lower) the quote price to rebalance their inventories. This is why the coefficients of contemporaneous order imbalance are positive in 5,10, and 15 minutes intervals. Then, if market makers think the pressure of inventories is not big enough in next period, they reverse the quote price. This is the reason that the coefficients of lagged-one order imbalance are negative in 5 and 15 minutes intervals. From 5, 10, and 15 minutes interval, the significantly positive coefficients of contemporaneous imbalance fall

from 50.73% to 40.21%, 52.58% to 34.02%, and 44.33% to 18.56%, at the 10%, 5%, and 1% significant level, respectively.

3.3 Dynamic Volatility-Order Imbalance GARCH(1,1) Relation

Table 5 presents dynamic relation between volatility of stock price and order imbalance. We expect that the larger the order imbalance, the higher the volatility. Though, the impact of the imbalance on volatility is not as strong as we have expected. At the 5% significant level, 14.43%, 17.53%, and 11.34% of order imbalance variables have significant positive impact on volatility for 5, 10, and 15 minutes interval respectively. At the 1% significant level, 8.25%, 6.19%, and 4.12% of order imbalance variables have significant positive impact on volatility for 5, 10, and 15 minutes interval respectively. The positive relation between volatility and order imbalance implies that the higher the order imbalance is, the more uncertain the return is.

The insignificant impact of imbalance on volatility indicates that market makers have the ability to mitigate volatility of SEO firms in announcements. In other words, market makers do have the capacity for stabilizing the price volatility of SEO firms in secondary market making.

3.4 Order Imbalance Based Trading Strategy

3.4.1 Trading strategy under the basis of quote price

We develop an order imbalance based trading strategy to test SEO market efficiency. We trim off 90% small imbalances because larger imbalances from discretionary investors tend to have much more substantial impact on returns. For each stock, we buy at ask when imbalance is positive and sell at bid when it turns negative. The result is listed in Panel A of Table 6.

Under the basis of quote price, the daily returns are -2.13%, -1.67%, and -1.53% for 5, 10 and 15 minutes interval respectively. The p-values are 0.310291, 0.491311, and 0.211818

respectively for 5, 10, and 15 minutes intervals, which reject the null. Then, we use a paired t-test in section 2 of Panel B to test whether our strategy can beat the original open-to-close return. The results show that the strategy fails. It implies an efficiency in season equity offerings market. We also investigate the difference of return among the three intervals. The result listed in section 3 of Panel B, we find that there is a significant difference between 10 minutes strategy and 5 minutes strategy.

3.4.2 Trading strategy under the basis of trade price

The trading returns are 0.07%, -3.11111E-05, and -0.17% for 5, 10, and 15 minutes intervals respectively when the trading strategy is under the basis of trade price. Again, we examine whether trading based on trade prices makes profits. We find that the strategy based on trade price earn significantly positive profits. In addition, we examine whether imbalance based trading strategy under the basis of trade price beats the original open-to-close return. In Section 2 of Panel B, we find that this strategy beat the original open-to-close return.

Finally, we investigate whether this strategy brings significantly different profits among 5, 10, and 15 minutes intervals. The empirical results show that there is no significant difference among the three intervals.

3.5. *The causal relationship in explaining return-order imbalance relationship*

To explore the dynamic return-order imbalance relationship during the price formation process, we employ a nested causality approach. In order to investigate the dynamic relationship between two variables, we impose the constraints in the upper panel of Table 1 for the VAR model. In Table 8, we present the empirical results of the tests of the hypotheses for the dynamic relationship in Table 2. Panel A presents the results for the entire sample. For the entire sample, we show that the unidirectional relationship from returns to order

imbalances is 6.19% of the sample firms for the entire sample, while the unidirectional relationship from order imbalances to returns is 9.28%. The percentage of firms that fall into the independent category is 36.08%. Moreover, 46.39% of firms exhibit a contemporaneous relationship between returns and order imbalances. Finally, 2.06% of firms exhibit a feedback relationship between returns and order imbalances. The percentage of firms exhibiting a unidirectional relationship from order imbalances to returns is larger than that exhibiting such a unidirectional relationship from returns to order imbalances, suggesting that order imbalances do constitute a better indicator for predicting future returns. This finding is consistent with many articles, which document that future daily returns could be predicted by daily order imbalances (Brown, Walsh, and Yuen, 1997; Chordia et al., 2004). In addition, the percentage of firms exhibiting a contemporaneous relationship is about twenty times that of the corresponding percentage reflecting a feedback relationship, indicating the interaction between returns and order imbalances in the current period is larger than that over the whole period.

In order to provide the evidence showing the impact on the relationship between returns and order imbalances, in Panel B we divide firms into three groups according to the firm size. We then test the multiple hypotheses of the relationship between returns and order imbalances. The results in Panel B indicate that the unidirectional relationship from order imbalances to returns is 15.63% for the small firm size quartiles, while the corresponding

percentage is 6.25% for the large firm size quartile during the entire sample period. The size-stratified results indicate that order imbalances constitute a better indicator for predicting future returns in smaller size quartile.

4. Conclusion

We choose 97 samples in 2004 that announce seasoned equity offering to study convergence process. First of all, we find that a significantly positive relation between lagged-one imbalances and return. Conditional on contemporaneous order imbalance, there is a negative relation between lagged-one imbalances and return. This is also consistent with Chordia and Subrahmanyam (2004).

We find that the relation between volatility and order imbalances is not strong. It implies that market makers have the capacity for stabilizing the price volatility of SEO firms. We also find a declining pattern as the interval increases.

Moreover, we develop an imbalance based trading. We use quote and trade prices to implement our strategy. The results show that only trade price basis earns a positive return, and beats the original open-to-close return. Finally, we employ a nested causality approach to examine the dynamic return-order imbalance relationship during the price-formation process. The results explain our imbalance-based trading strategy.

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Table 1 Hypotheses on the dynamic relationship of a bivariate system

The bivariate VAR model may be expressed as :
$$\begin{bmatrix} \phi_{11}(L) & \phi_{12}(L) \\ \phi_{21}(L) & \phi_{22}(L) \end{bmatrix} \begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} = \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$
 where x_{1t}

and x_{2t} are mean adjusted variables. The first and second moments of the error structure,

$\varepsilon_t = (\varepsilon_{1t}, \varepsilon_{2t})'$ are that $E(\varepsilon_t) = 0$, and $E(\varepsilon_t \varepsilon_{t+k}) = 0$ for $k \neq 0$ and $E(\varepsilon_t \varepsilon_{t+k}) = \Sigma$

for $k=0$, where $\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$

The causal relationships are defined as follows: $\wedge$ represents independency; $< - >$ is the contemporaneous relationship; $\neq >$ is the negation of a unidirectional relationship; $< = >$ is the feedback relationship; $\neq > >$ is the negation of a strong unidirectional relationship where $\sigma_{12} = \sigma_{21} = 0$; and $< < = > >$ is a strong feedback relationship where $\sigma_{12} = \sigma_{21} = 0$.

Hypotheses	The VAR test
$H_1 : x_1 \wedge x_2$	$\phi_{12}(L) = \phi_{21}(L) = 0$, and $\sigma_{12} = \sigma_{21} = 0$
$H_2 : x_1 < - > x_2$	$\phi_{12}(L) = \phi_{21}(L) = 0$
$H_3 : x_1 \neq > x_2$	$\phi_{21}(L) = 0$
$H_3^* : x_2 \neq > x_1$	$\phi_{12}(L) = 0$
$H_4 : x_1 < = > x_2$	$\phi_{12}(L) * \phi_{21}(L) \neq 0$
$H_5 : x_1 \neq > > x_2$	$\phi_{21}(L) = 0$, and $\sigma_{12} = \sigma_{21} = 0$
$H_6 : x_2 \neq > > x_1$	$\phi_{12}(L) = 0$, and $\sigma_{12} = \sigma_{21} = 0$
$H_7 : x_1 < < = > > x_2$	$\phi_{12}(L) * \phi_{21}(L) \neq 0$, and $\sigma_{12} = \sigma_{21} = 0$

Table 2 Test flow chart of a multiple hypothesis testing procedure

Five groups of dynamic relationship are identified: independency ($\wedge$) the contemporaneous relationship ($\leftrightarrow$) the unidirectional relationship ($\Rightarrow$ or $\Leftarrow$) and feedback relationship ($<=>$). To determine a specific causal relationship, we use a systematic multiple hypotheses testing method. Unlike the traditional pairwise hypothesis testing, this testing method avoids the potential bias induced by restricting the causal relationship to a single alternative hypothesis. In implementing this method, we need to employ results of several pairwise hypothesis tests. For instance, in order to conclude that $x_1=>x_2$, we need to establish that $x_1<\neq x_2$ and to reject $x_1\neq>x_2$. To conclude that $x_1<->x_2$, we need to establish that $x_1<\neq x_2$ as well as $x_1\neq>x_2$ and also to reject $x_1\wedge x_2$. In other words, it is necessary to examine all five hypotheses in a systematic way before a conclusion regarding the dynamic relationship can be drawn.

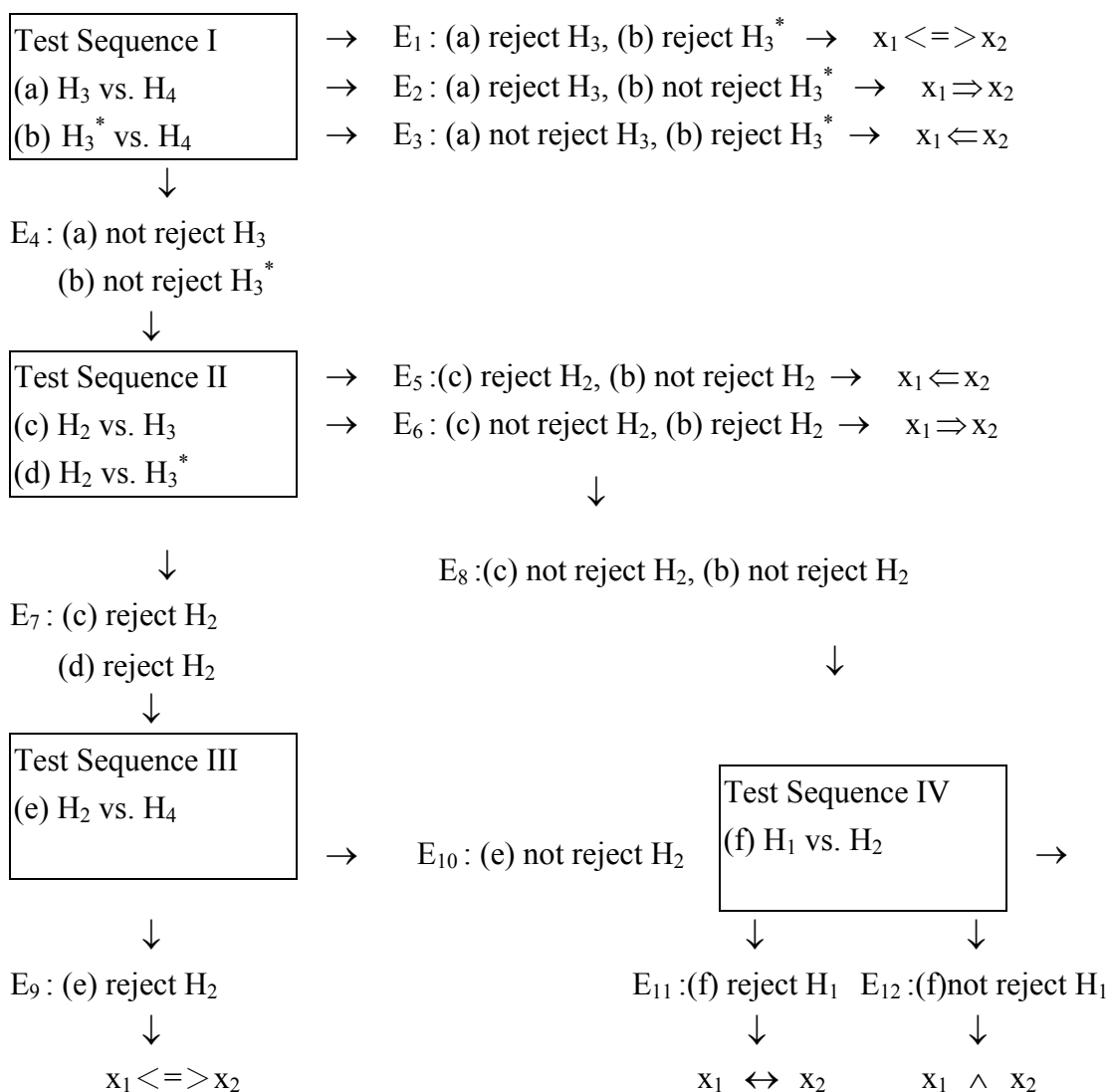


Table 3 Empirical Results of the Unconditional Lagged Return-Order Imbalance OLS Relation

$$R_t = \alpha_0 + \alpha_1 OI_{t-1} + \alpha_2 OI_{t-2} + \alpha_3 OI_{t-3} + \alpha_4 OI_{t-4} + \alpha_5 OI_{t-5} + \varepsilon_t$$

where R_t is the current stock return of the individual stock on time t ,

OI_{t-i} , $i=1, 2, 3, 4$, and 5 , are lagged order imbalance at time $t-1, t-2, t-3, t-4$, and $t-5$ of each individual stock,

$\alpha_0, \alpha_i, i=1, 2, 3, 4$, and 5 , are intercepts and coefficients of the

corresponding lagged order imbalance,

and ε_t is the residual of the current stock return.

“Significant” denotes significance at the 5% level.

Panel A: 5-minute interval

	Average coefficient	Positive and significant	Negative and significant
OI_{t-1}	-1.02E-08	3.09%	4.12%
OI_{t-2}	1.56E-08	4.12%	4.12%
OI_{t-3}	-5.60E-09	5.15%	10.31%
OI_{t-4}	1.28E-09	3.09%	3.09%
OI_{t-5}	-1.01E-08	2.06%	3.09%

Panel B: 10-minute interval

	Average coefficient	Positive and significant	Negative and significant
OI_{t-1}	9.65E-09	6.19%	4.12%
OI_{t-2}	-5.88E-09	0.00%	4.12%
OI_{t-3}	-2.52E-08	3.09%	2.06%
OI_{t-4}	9.69E-09	1.03%	4.12%
OI_{t-5}	-1.00E-08	1.03%	3.09%

Panel C: 15-minute interval

	Average coefficient	Positive and significant	Negative and significant
OI_{t-1}	-2.17E-08	1.03%	4.12%
OI_{t-2}	9.05E-11	3.09%	1.03%
OI_{t-3}	7.61E-09	3.09%	3.09%
OI_{t-4}	-2.71E-08	3.09%	2.06%
OI_{t-5}	1.07E-08	2.06%	1.03%

Table 4 Empirical Results of the Conditional Contemporaneous Return-Order Imbalance OLS Relation

$$R_t = \alpha_0 + \alpha_1 OI_t + \alpha_2 OI_{t-1} + \alpha_3 OI_{t-2} + \alpha_4 OI_{t-3} + \alpha_5 OI_{t-4} + \varepsilon_t,$$

where R_t is the current stock return of the individual stock on time t ,

OI_{t-i} , $i=0, 1, 2, 3$, and 4 , are contemporaneous order imbalance at time t and lagged order imbalance at time $t-1$, $t-2$, $t-3$, and $t-4$ of each individual stock,

α_0 , α_i , $i=1, 2, 3, 4$, and 5 , are intercepts and coefficients of the

corresponding contemporaneous and lagged order imbalance,

and ε_t is the residual of the current stock return.

“Significant” denotes significance at the 5% level.

Panel A: 5-minute interval

	Average coefficient	Positive and significant	Negative and significant
OI_t	1.12E-07	52.58%	3.09%
OI_{t-1}	-1.24E-08	2.06%	5.15%
OI_{t-2}	1.94E-08	4.12%	4.12%
OI_{t-3}	-4.88E-09	4.12%	10.31%
OI_{t-4}	1.06E-09	3.09%	6.19%

Panel B: 10-minute interval

	Average coefficient	Positive and significant	Negative and significant
OI_t	1.00E-07	34.02%	3.09%
OI_{t-1}	9.27E-09	7.22%	6.19%
OI_{t-2}	4.94E-09	0.00%	3.09%
OI_{t-3}	-2.11E-08	3.09%	3.09%
OI_{t-4}	1.79E-09	1.03%	2.06%

Panel C: 15-minute interval

	Average coefficient	Positive and significant	Negative and significant
OI_t	1.22E-07	34.02%	4.12%
OI_{t-1}	-2.34E-08	4.12%	5.15%
OI_{t-2}	-4.52E-09	3.09%	1.03%
OI_{t-3}	1.32E-08	4.12%	1.03%
OI_{t-4}	-1.65E-08	1.03%	3.09%

Table 5 Empirical Results of Dynamic Volatility-Order Imbalance GARCH(1,1)
Relation

$$R_t = \alpha + \varepsilon_t$$

$$\varepsilon_t | \Omega_{t-1} \sim N(0, h_t)$$

$$h_t = A + Bh_{t-1} + C\varepsilon_{t-1}^2 + \gamma * OI_t$$

where R_t is the return in period t , and is defined as $\ln\left(\frac{P_t}{P_{t-1}}\right)$,

OI_t is the explanatory variable, order imbalance,

γ is the coefficient describing the impact of order imbalance on stock volatility,

ε_t is the residual value of the stock return in period t ,

Ω_{t-1} is the information set in period $t-1$,

and α , A , B , and C are intercepts and coefficients.

“Significant” denotes significant at the 5% level.

	Percent positive and significant	Percent negative and significant
5-min interval	14.43%	4.12%
10-min interval	17.53%	1.03%
15-min interval	11.34%	2.06%

Table 6 Hypothesis test of trading profit under the basis of quote price

Panel A: Returns compared with zero

$$\begin{cases} H_0: \mu_i \leq 0 \\ H_1: \mu_i > 0 \end{cases}$$

where μ_i is the return of the trading strategy

i denotes 5-, 10-, and 15-min intervals

	Mean	P-value
5-min return strategy	-0.0214	0.310291
10-min return strategy	-0.0167	0.491311
15-min return strategy	-0.0153	0.211818

Panel B: Returns compared with returns of buy-and-hold strategy

$$\begin{cases} H_0: \mu_i \geq \mu_0 \\ H_1: \mu_i < \mu_0 \end{cases}$$

where μ_i is the return of the trading strategy, μ_0 is the original open-to-close return

i denotes 5-, 10-, and 15-min intervals

	Mean	P-value
Original open-to-close return	-0.0074	
5-min return strategy	-0.0214	4.9965E-05
10-min return strategy	-0.0167	0.0302
15-min return strategy	-0.0153	0.0313

Panel C: Differences in returns among the three intervals

$$\begin{cases} H_0: \mu_i = \mu_j \\ H_1: \mu_i \neq \mu_j \end{cases}$$

where μ_i is the return of the trading strategy

i, j denotes 5-, 10-, and 15-min intervals, $i \neq j$

P-value	5-min return	10-min return	15-min return
5-min return			
10-min return	0.0321		
15-min return	0.2489	0.7804	

Table 7 Trading Profit under the Basic of Trade Price

Panel A: Returns compared with zero

$$\begin{cases} H_0: \mu_i \leq 0 \\ H_1: \mu_i > 0 \end{cases}$$

where μ_i is the return of the trading strategy
i denotes 5-, 10-, and 15-min intervals

	Mean	P-value
5-min return strategy	-0.0214	5.0226E-09
10-min return strategy	-0.0167	4.3153E-06
15-min return strategy	-0.0153	5.0252E-06

Panel B: Returns compared with returns of buy-and-hold strategy

$$\begin{cases} H_0: \mu_i \geq \mu_0 \\ H_1: \mu_i < \mu_0 \end{cases}$$

where μ_i is the return of the trading strategy, μ_0 is the original open-to-close return
i denotes 5-, 10-, and 15-min intervals

	Mean	P-value
Original open-to-close return	-0.0074	
5-min return strategy	0.00074	0.9997
10-min return strategy	-3.1111E-05	0.9988
15-min return strategy	-0.0017	0.9348

Panel C: Differences in returns among the three intervals

$$\begin{cases} H_0: \mu_i = \mu_j \\ H_1: \mu_i \neq \mu_j \end{cases}$$

where μ_i is the return of the trading strategy
i, j denotes 5-, 10-, and 15-min intervals, $i \neq j$

P-value	5-min return	10-min return	15-min return
5-min return			
10-min return	0.9622		
15-min return	0.5796	0.0851	

Table 8 Dynamic nested causality relationship between returns and order imbalances

The causal relationships are defined as follows: $\wedge$ represents independency; $< - >$ is the contemporaneous relationship; $\neq >$ is the negation of the unidirectional relationship; $< = >$ is the feedback relationship; $\neq > >$ is the negation of a strong unidirectional relationship where $\sigma_{12} = \sigma_{21} = 0$; and $< < = > >$ is a strong feedback relationship where $\sigma_{12} = \sigma_{21} = 0$.

	$x_1 \wedge x_2$	$x_1 < - > x_2$	$x_1 \Rightarrow x_2$	$x_1 \Leftarrow x_2$	$x_1 < = > x_2$
Panel A: All size					
All trade size	36.08%	46.39%	6.19%	9.28%	2.06%
 Panel B: Firm size					
Small firm size	25.00%	53.13%	3.13%	15.63%	3.13%
Medium firm size	45.45%	45.45%	3.03%	6.06%	0.00%
Large firm size	37.50%	40.63%	12.50%	6.25%	3.13%